

CMCP Evidence Accounting Beyond Linear Score Space

Experimental findings, Schweizer–Sklar, and an information-geometric proposal

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Abstract

CMCP is an evidence ledger for test-time constraint packets. Its original weight is a Schur-complement residual in a score-vector representation. Experiments establish useful accounting and replay-budget effects, but reveal a central limitation: a provenance-distinct systematic bit-flip can be almost fully novel in that linear score space. Distance-correlation novelty plus a Schweizer–Sklar t-norm suppresses this constructed bias, but calibration selects the product limit rather than a non-product t-norm. This review separates those observations from a proposed conditional distributional increment based on alpha divergence, with Fisher–Rao/Hellinger as the symmetric primary. The proposal is falsifiable; it is not established conditional mutual information or a guaranteed solution.

1 Problem and current CMCP

CMCP packets contain a batch-aligned constraint-value vector, mechanism family, provenance/innovation identifiers, and a precision weight. Its role is to decide how much extra statistical weight a packet deserves given a ledger of stored packets. In reduced E4, this weight controls direct-logit constraint settlement and, in the parameter-learning follow-up, replay strength.

For a candidate packet j , let $s_j \in \mathbb{R}^d$ be its standardized score residual. With stored scores $S = \{s_i\}$, the original conditional ratio is

$$r_{\text{score}}(j \mid S) = \frac{F_{jj} - F_{jS}(F_{SS} + \lambda I)^\dagger F_{Sj}}{F_{jj}}, \quad F_{ab} = d^{-1} s_a^\top s_b, \quad (1)$$

clipped to $[0, 1]$. It is zero for a score explained by the stored linear span and near one for an orthogonal score. A prior provenance gate selects one stable representative per innovation identifier.

This criterion has genuine strengths: exact duplicate/descendant idempotence, retention of independent repeated evidence, and tested equivariance under supplied orthogonal score-coordinate alignment. But it is a linear conditional-energy criterion on selected features. It is neither general conditional mutual information nor a test for arbitrary deterministic dependence.

2 What was observed

2.1 Accounting and persistent bursts

Five held-out single-shot confirmation seeds gave exact idempotence for exact duplicates and deterministic descendants, mean weight 0.997862 for an independent second packet, and order invariance after canonicalization. Partial redundancy was high: 0.657531 versus constructed oracle 0.5. Naively counting a duplicate twice slightly improved immediate loss; the initial result was accounting, not performance evidence.

The corrected eight-episode persistent experiment reused one explicitly identified cohort and changed extraction noise by episode. An earlier fresh-batch smoke was rejected because row-aligned packets were applied to unrelated examples. On five sealed confirmation seeds at episode 8:

Metric	CMCP	oracle	naive accumulation
Effective precision	8.89433	9.0	26.0
Task accuracy	0.83281	—	0.74844
ECE (10-bin)	0.07797	—	0.12309
Multiclass Brier score	0.22596	—	0.32489

Observed. CMCP prevented duplicate count from becoming precision inflation and improved same-cohort accuracy/calibration in this frozen-weight protocol.

2.2 Parameter learning and replay

A typed store now carries cohort and ordered-example identity, packet/provenance/innovation identity, mechanism family, precision, and values hash. It fails closed on row permutation, cohort mismatch, duplicate packet IDs, and conflicting innovation payloads. In a three-seed six-episode reversal, CMCP precision was 6.8364–7.9343 against oracle 7.5, while naive reached 21.0. Yet naive retained Task A better on every seed and usually matched Task-B learning; five confirmation seeds were not opened.

A replay-coefficient sweep produced a bounded plasticity signal. Increasing coefficient from 0.15 to 0.30 to 0.50 moved mean CMCP/naive Task-B crossover from episode 6 to 5 to 3. At episode 6, CMCP minus naive Task-B accuracy was +0.031, +0.099, and +0.133. This supports CMCP as a replay-budget normalizer under stronger replay pressure. It does not establish a universal continual-learning win; naive had lower ECE in this exploratory sweep.

2.3 The technical failure

In a provenance-distinct systematic bit-flip smoke, raw score-space ratio was 0.9461: nearly fully novel under Equation 1. Thus original score-space CMCP did not recognize the deterministic transformation. Earlier zero weight in a related test was explained by shared provenance, not by the Schur test. A deterministic systematic transformation can look linearly independent in chosen score features.

3 Schweizer–Sklar result

The prototype used raw-evidence novelty

$$r_{\text{ev}}(j \mid S) = 1 - \max_{i \in S} \text{dCor}(v_j, v_i), \quad (2)$$

where dCor is the implemented finite-sample biased Szekely–Rizzo distance correlation, averaged factorwise. It combines score and evidence novelty by

$$T_p(x, y) = [\max\{0, x^p + y^p - 1\}]^{1/p}, \quad p > 0, \quad r_Q = T_p(r_{\text{score}}, r_{\text{ev}}). \quad (3)$$

As $p \rightarrow 0^+$, this tends to product xy ; $p = 1$ is the Lukasiewicz t-norm.

On three calibration seeds with $p \in [0.05, 1.5]$, selected endpoint was $p = 0.05$:

Fixture	exact duplicate	bit-flip	independent	50% overlap
Mean r_Q	0.000	0.000	0.870930	0.363021

The selection score fell monotonically from 0.833950 at $p = 0.05$ to 0.553181 at $p = 1.5$. An auxiliary product-limit computation ($p = 0.001$) was slightly better on the same constructed objective. A provenance-distinct bit-flip smoke assigned 1.0/0.0 to original/biased packets with model weights unchanged.

Observed. Evidence dependence catches this constructed complement relation. The data supports a product-like score/evidence combination, not a specifically non-product Schweizer–Sklar geometry: the optimum is at a tested boundary. Biased finite-sample dCor and partial-overlap target 0.4 are not independently calibrated truths.

4 Why not standard MI?

Mutual information requires a specified joint distribution over packets, targets/parameters, and conditioning ledger. Equation 1 is a useful Gaussian-linear analogue, not general MI. A neural MI estimator would introduce a learned critic, substantial sample requirements, and a new route for fixture overfit.

A more disciplined next object is an *information-geometric distributional increment*: assess packets by the predictive distributions they induce on a fixed typed cohort. This is a testable estimator, not conditional MI unless a posterior or generative model makes that identification valid.

5 Proposed math: conditional distributional novelty

Fix a cohort C with ordered example identity. Let $P_0(c)$ be feed-forward categorical prediction, $P_S(c)$ prediction after stored packets S , $P_j(c)$ after candidate j alone, and $P_{S \cup \{j\}}(c)$ after ledger plus candidate. All four are computed by frozen-weight direct-logit settlement.

Let $\bar{D}_\alpha$ be cohort-mean Amari alpha divergence:

$$D_\alpha(P||Q) = \frac{4}{1-\alpha^2} \left[1 - \sum_k P_k^{(1-\alpha)/2} Q_k^{(1+\alpha)/2} \right], \quad -1 < \alpha < 1. \quad (4)$$

Its endpoints are directed KL divergences. At $\alpha = 0$, $D_0 = 4(1 - \sum_k \sqrt{P_k Q_k})$, four times squared Hellinger distance. This symmetric case is bounded for categorical predictions and locally induces the Fisher metric, making it a sound primary geometry.

Define operational conditional novelty by

$$r_\alpha^{\text{geom}}(j | S) = \text{clip}_{[0,1]} \frac{\bar{D}_\alpha(P_S || P_{S \cup \{j\}})}{\bar{D}_\alpha(P_0 || P_j) + \epsilon}. \quad (5)$$

The numerator is residual predictive displacement after the ledger; denominator is candidate stand-alone displacement. No additional settled predictive change gives zero; disjoint effects should give a ratio near one.

For a kernel formulation define alpha affinity

$$K_\alpha(P, Q) = |C|^{-1} \sum_{c \in C} \sum_k P_k(c)^{(1-\alpha)/2} Q_k(c)^{(1+\alpha)/2}. \quad (6)$$

At $\alpha = 0$, this is Bhattacharyya/Hellinger affinity: a positive-definite kernel on square-root probability vectors and directly Fisher–Rao motivated. Nonzero alpha is best treated as a directed divergence family, not casually as a symmetric kernel.

Hypothesis. A bit flip can be linearly orthogonal in present score features yet produce little residual predictive displacement after source constraint is active. Equation 5 would then be low where Schur residualization was 0.9461. Unlike raw dCor, it tests the model response to the packet intervention.

This is not guaranteed. If bit-flip constraint genuinely moves predictions in a distinct direction, Fisher–Rao/alpha geometry should report novelty despite deterministic raw relation. Then suitable policy may require both typed provenance/dependence and geometric predictive increment. The proposal must therefore face counterexamples and held-out confirmation.

6 Falsification-first protocol

1. **Estimator audit:** stable log-space alpha divergence, endpoint KL limits, Hellinger primary, and permutation or unbiased dependence baselines. Test non-negativity, identity, zero-probability stability, cohort-order invariance, frozen weights.
2. **Broaden fixtures:** exact copies, bit flips, factor permutations, cross-factor deterministic maps, affine/logit maps, noisy copies across SNR, and independent packets.
3. **Compare estimators:** linear Schur, product+dCor, Hellinger ratio, alpha grid. Calibration selects primary and thresholds; product remains a baseline.
4. **Seal confirmation:** fresh seeds and fresh transform mixtures; require both discrimination and downstream replay-budget plasticity at fixed update counts.
5. **Defer learning:** any learned copula or critic trains on a separate fixture-generating distribution and is evaluated on held-out mechanism families while retaining Hellinger/product baselines and typed-cohort contracts.

7 Questions for reviewers

1. Is settled predictive displacement the right object, or must packet information first be defined by a posterior or likelihood model?
2. Should primary geometry be symmetric Hellinger/Fisher–Rao ($\alpha = 0$), or directed KL because updates are asymmetric?
3. Does direct-logit settlement define a meaningful intervention, or should comparison live in parameter, natural-gradient, or belief space?
4. Which invariances should a kernel respect: factor relabeling, known rotations, deterministic invertible maps, cohort permutation?
5. Can this ratio distinguish useful correlated repetition from redundant repetition without silently preferring novelty?
6. What held-out fixtures would rule out a learned kernel merely memorizing the present grammar?

8 Current decision

Proposal. Do not replace CMCP in the main path yet. Retain provenance deduplication and product-like dCor as exploratory baselines. Implement Equation 5 as a separate preregistered estimator comparison with $\alpha = 0$ Hellinger primary; treat nonzero-alpha selection as empirical, not a consequence of information geometry. A learned kernel is premature until a broad held-out fixture distribution and reliable geometric baseline exist.

9 Evidence and references

- CMCP ledger confirmation: [experiments/20260725T001900Z-e4-cmcp-ledger-confirmation/RUN.md](#).
- Persistent same-cohort confirmation: [experiments/20260725T015835Z-e4-cmcp-persistent-confirmation/RUN.md](#).
- Typed retention calibration: [experiments/20260725T021200Z-e4-cmcp-typed-retention-calibration/RUN.md](#).
- Replay sweep: [experiments/20260725T031220Z-e4-cmcp-replay-sweep-final/RUN.md](#).

- Biased-injection smoke: `experiments/20260725T031257Z-e4-cmcp-biased-injection-smoke/RUN.md`.
- Schweizer–Sklar calibration and smoke: `experiments/20260725T034212Z-e4-quantale-calibration/RUN.md` and `20260725T034222Z-e4-quantale-biased-smoke/RUN.md`.
- S.-I. Amari, *Information Geometry and Its Applications*, Springer, 2016.
- G. J. Székely, M. L. Rizzo, N. K. Bakirov, Measuring and testing dependence by correlation of distances, *Annals of Statistics*, 2007.